Supplementary Material for *Inference, Not Just Optimisation: Streaming Inversion-Free Quasi-Newton Estimation on Dependent Data Streams*

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This Supplementary Material contains four figures. Each supports a claim that the manuscript also establishes by other means, so the manuscript is self-contained without them; they are collected here to keep the manuscript inside Information Sciences' suggested maximum of ten figures plus tables for a theoretical article. Theorem, proposition, equation, table and section numbers in the captions are those of the *manuscript*, resolved from it automatically rather than transcribed; figure numbers prefixed with S are local to this document. Every number in the captions is generated from the saved result files in the public repository, https://github.com/Kendiukhov/streaming-quasi-newton-dependent-data.

References

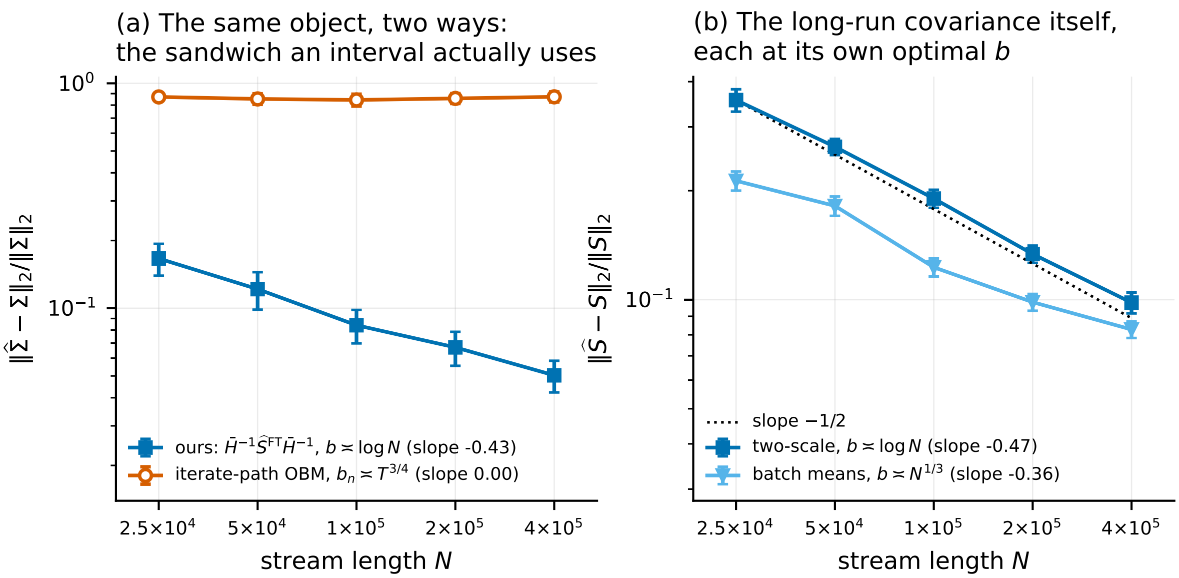

Figure S1: The rate claim, measured, on the object an interval actually uses. Both axes logarithmic; error bars are ± 1.96 Monte Carlo standard errors; fitted slopes in the legends. **(a)** Relative operator-norm error in estimating the whole sandwich $\Sigma = H^{-1}SH^{-1}$. Ours is $\bar{H}^{-1}\widehat{S}^{\mathrm{FT}}\bar{H}^{-1}$ at $b \asymp \log N$; the comparator is overlapping batch means on the *iterate* path at the block length $b_n \asymp T^{3/4}$ that analysis requires (Appendix C.3 gives the form), which uses no gradients and no curvature estimate. Same target, so the comparison is meaningful; but read the levels, not the slopes. The iterate-path curve is flat here, and its published $N^{-1/8}$ rate is too slow for our range to resolve, so its fitted slope says nothing about it. What the panel shows is that over the sample sizes we can reach, one estimator is accurate and the other is not. **(b)** Error in estimating S alone, each estimator at its own optimal block length: the two-scale form at $b \asymp \log N$ against the $-1/2$ of Theorem 9(iv), plain batch means at $b \asymp N^{1/3}$ against $-1/3$.

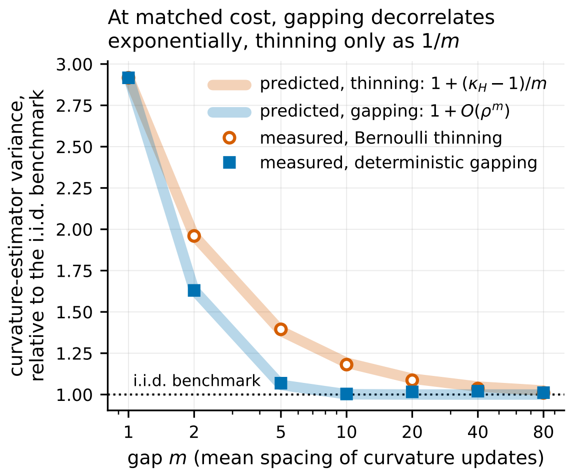

Figure S2: At matched cost, a deterministic gap decorrelates curvature updates exponentially in m while random thinning only does so linearly (Proposition 10). Vertical axis: $n\,\mathbb{E}\|\bar{H} - H\|_F^2$ divided by its exact independent-data value $(\operatorname{tr}\Sigma)^2 + \|\Sigma\|_F^2$ — a computed constant, not a fitted normalisation. Pale bands are the two predictions; markers are measurements with the optimisation frozen at $\theta^\star$, so only estimation quality is compared. $\kappa_H = (1+\phi^2)/(1-\phi^2)$ here.

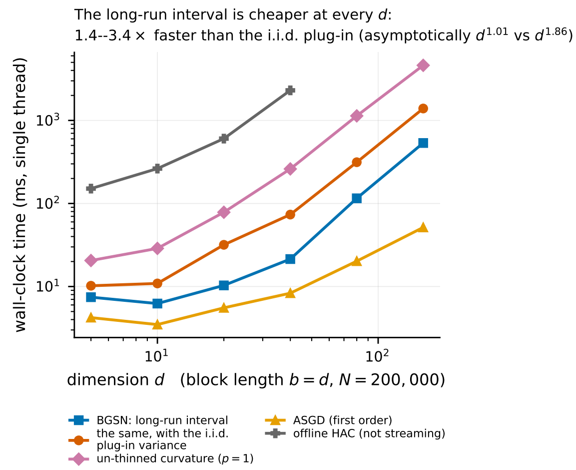

Figure S3: The statistically correct interval is the cheaper one. Wall-clock milliseconds for one pass over $N = 200{,}000$ observations (vertical axis, log scale) against dimension d (horizontal, log scale), single-threaded, minimum over seven interleaved repetitions. The i.i.d. plug-in score covariance costs $O(d^2)$ per *observation*; the blocked long-run covariance costs $O(d^2)$ per *block*. Slopes are fitted on the four largest d; at this N they are inflated by the one-off warm-up (see Eq. (8) and Table D.5, which also gives the asymptotic exponents), and the two smallest d are dominated by fixed per-block overhead rather than by either term of Eq. (8). The message of the figure is the vertical separation, which holds at every d.

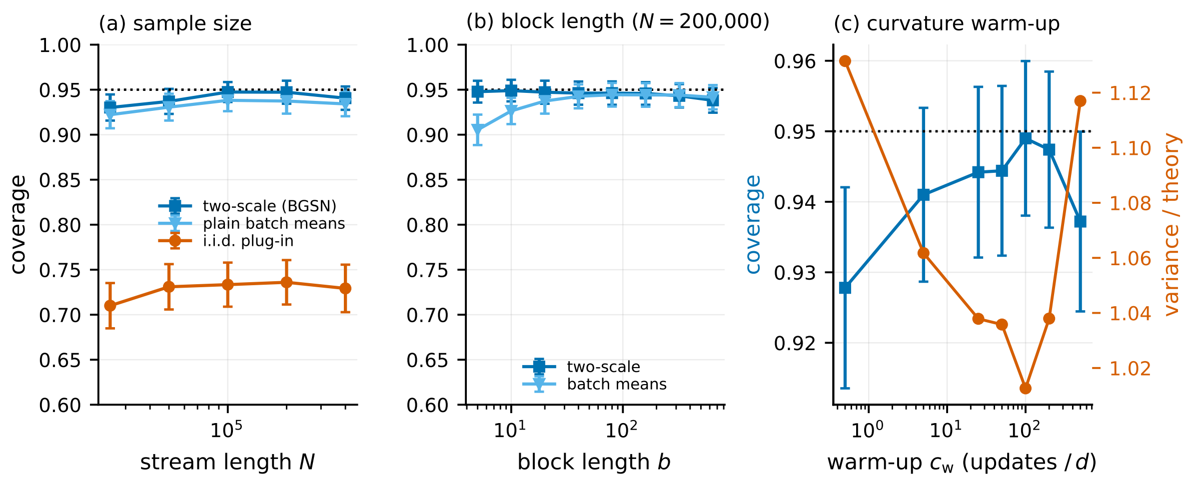

Figure S4: Ablations. (a) Coverage converges to nominal in N for the two-scale variance, while the dependence-blind interval converges to 0.749, the value Proposition 8 predicts for this design. (b) Coverage is flat in the block length for the two-scale variance and degrades at both ends for plain batch means — the practical content of the $O(\rho^b)$-versus-$O(1/b)$ distinction. (c) The curvature warm-up matters, though the step-length safeguard of Eq. (6) now absorbs part of its job: with essentially no warm-up coverage on this design is 0.928 against 0.949 at the default, and the relative variance on the right axis tells the same story. On the regime-switching design, where the warm-up is load-bearing rather than merely helpful, the effect is far larger (Table D.4(c′)).